Author: Oghenekevwe Jeffery Onodje

Title: Network Scan Report 1

Tool Used: Nmap and Command Terminal

Target: Localhost

Date: 26th March, 2026

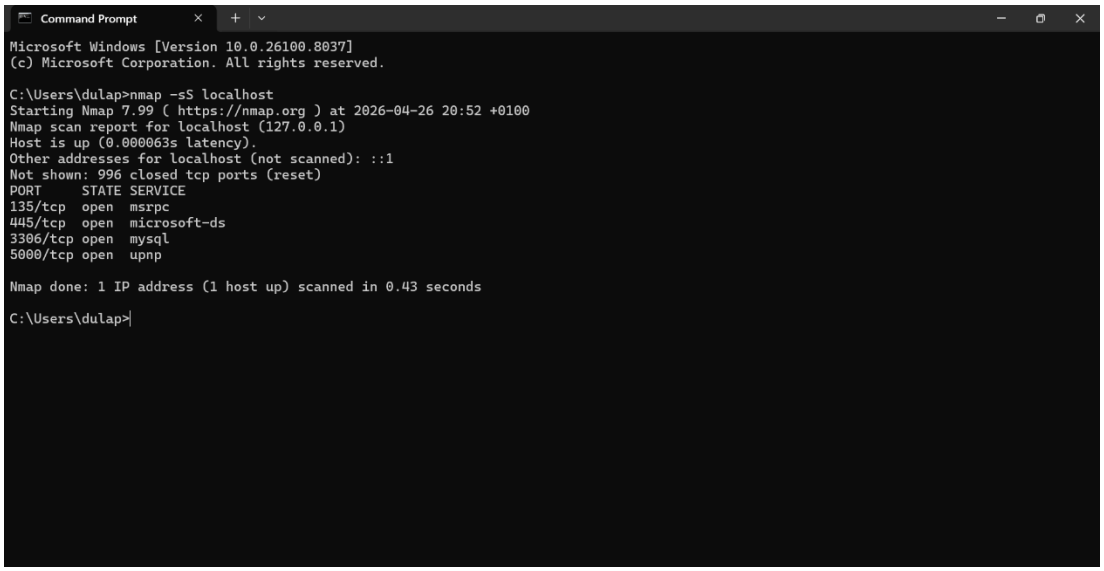
Objective: to perform a basic network scan on a local system using Nmap in order to identify open ports and running services, and to understand potential security risks.

METHODOLOGY:

- A TCP scan was carried out using the command line: **nmap -sS localhost**
  This is to discover open ports.
- A service version detection scan was also carried out using the command line: **nmap -sV localhost**
  This shows what software is running on an open port and its version.
- A host discovery scan was then carried out using the command line: **nmap -sn 192.168.1.0/24**
  This is simply to discover what devices are active on the network

EVIDENCE:

1. **Figure 1: Nmap TCP SYN Scan Result**
   The output shows multiple open ports on the localhost network.



2. **Figure 2: Service Version Scan Result**

```
Command Prompt

C:\Users\dulap>nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-26 21:20 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00010s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE       VERSION
135/tcp  open  msrpc         Microsoft Windows RPC
445/tcp  open  microsoft-ds?
3306/tcp open  mysql         MySQL 8.0.43
5000/tcp open  http          Node.js (Express middleware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.48 seconds

C:\Users\dulap>
```

3. **Figure 3: Host Discovery Scan Result**

```
Command Prompt

C:\Users\dulap>nmap -sn 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-26 21:24 +0100
Nmap done: 256 IP addresses (0 hosts up) scanned in 207.42 seconds

C:\Users\dulap>
```

FINDINGS:

Port	State	Service	Description	Risk
135	Open	msrpc	Microsoft Windows RPC	Medium - High
445	Open	Microsoft-ds	SMB	High
3306	Open	mysql	MySQL 8.0.43	Medium
5000	Open	http	Node.js(Express Middleware)	Medium - High

Further enumeration could be performed on detected services to identify vulnerabilities.

SECURITY ANALYSIS:

- The scan identified multiple open ports, including 115 (MSRPC), 445 (SMB), 3306 (MySQL), and 5000 (HTTP).
- Port 445 (SMB) presents a critical risk due to its history of exploitation in attacks such as the WannaCry ransomware. Immediate restriction is recommended using a firewall.
- Port 3306 (MySQL) exposes a database service which may be vulnerable to brute-force or unauthorized access if not properly secured. Remedy: Enforce strong authentication on MySQL.
- Port 5000 indicates a Node.js web application, which may introduce application-level vulnerabilities if not hardened. Remedy: Ensure that the NodeJS Application is not publicly exposed.
- Overall, the system presents a moderate to high attack surface and requires immediate hardening, particularly around SMB exposure and database access controls.

LIMITATIONS:

This scan was conducted on localhost, meaning results may differ in a real-world network environment. External exposure and firewall configurations were not assessed.

CONCLUSION:

The scan identified several open ports and active services that increase the system's attack surface. High-risk services such as SMB, along with exposed database and web application ports, may be exploited if not properly secured. Implementing access controls, restricting unnecessary services, and applying security best practices are essential to reduce potential vulnerabilities.

FUTURE WORK:

- Perform vulnerability scanning using tools like Nessus
- Conduct deeper service enumeration
- Analyze web application security on port 5000